2. Measure

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Definition 2.1: A *null set* $A \subseteq \mathbb{R}$ is a set that may be covered by a sequence of intervals of arbitrarily small total length, i.e. given any $\epsilon > 0$ we can find a sequence $\{I_n : n \geq 1\}$ of intervals such that

$$A \subseteq \bigcup_{n=1}^{\infty} I_n$$

and

$$\sum_{n=1}^{\infty} l(I_n) < \epsilon.$$

(We also say simply that *'A is null'*.)

Theorem 2.2. *If* $(N_n)_{n\geq 1}$ *is a sequence of null sets, then their union*

$$N = \bigcup_{n=1}^{\infty} N_n$$

is also null.

Crux. The $\epsilon/2^n$ **trick**. For each N_n, find a covering $\{I_k^n\}_{k\geq 1}$ so that $\sum_{k=1}^{\infty} l(I_k^n) < \epsilon/2^n$.

Collectively, the countable collection $\{I_k^n\}_{n\geq 1, k\geq 1}$ covers N. Because the lengths are non-negative, the double summation commutes (arrangement doesn't matter), yielding a countable covering whose total length is bounded by:

$$\sum_{n=1}^{\infty} \sum_{k=1}^{\infty} l(I_k^n) < \sum_{n=1}^{\infty} \frac{\epsilon}{2^n} = \epsilon$$

$\square$

Definition 2.3: The (Lebesgue) *outer measure* of any set $A \subseteq \mathbb{R}$ is given by

$$m^*(A) \triangleq \inf Z_A$$

where

$$Z_A \triangleq \left\{ \sum_{n=1}^{\infty} l(I_n) : \; I_n \text{ are intervals, } A \subseteq \bigcup_{n=1}^{\infty} I_n \right\}$$

Note that I_n can be empty so that even finite covers are allowed in the definition of Z_A.

Theorem 2.4. *$A \subseteq \mathbb{R}$ is a null set if and only if $m^*(A) = 0$.*

Crux. Recall $m^*(A) = \inf Z_A$, where Z_A is the set of all covering length sums.

- ($\Leftarrow$): If $m^*(A) = 0$, then $\forall \epsilon > 0$, we can find a covering sum in Z_A that is $< \epsilon$. This is exactly the definition of a null set.
- ($\Rightarrow$): If A is null, then $\forall \epsilon > 0$, there exists a cover of length $< \epsilon$, meaning there are arbitrarily small positive numbers in Z_A. Therefore, $m^*(A) = \inf Z_A = 0$.

□

Proposition 2.5 (Monotonicity) If $A \subseteq B$, then $m^*(A) \leq m^*(B)$.

Crux. Every cover of B is automatically a cover of A. Therefore, the set of all covering sums satisfies $Z_B \subseteq Z_A$. Since taking the infimum reverses the direction of set inclusion ($\inf Z_A \leq \inf Z_B$), the result $m^*(A) \leq m^*(B)$ immediately follows. □

Theorem 2.6. *The outer measure of an interval equals its length.*

Crux. Let $I = [a, b]$.

- **Upper Bound** ($m^*(I) \leq l(I)$): Trivial. For any $\epsilon > 0$, the open interval $(a-\epsilon, b+\epsilon)$ covers I, so $m^*(I) < b - a + 2\epsilon$.
- **Lower Bound Setup (Heine-Borel):** Take an arbitrary cover $\{I_n\}$ such that $\sum l(I_n) < m^*(I) + \epsilon/2$. Expand each I_n into an open interval J_n by adding $\epsilon/2^{n+1}$ length. Since $[a, b]$ is compact, **Heine-Borel** guarantees a *finite* open subcover $\{J_1, \ldots, J_m\}$.
- **ERRATA - The Overlap Argument:** The text sloppily claims $b - a < \max(d_n) - \min(c_n) \leq \sum l(J_n)$ without properly justifying the overlap. (Confirmed on MSE: SE5122341).

- **Correction (The Chain Algorithm):** Discard redundant intervals. Order the remaining finite cover left-to-right: $J_1 = (c_1, d_1)$ covers a, $J_2 = (c_2, d_2)$ covers d_1, etc., until J_k covers b. Because they overlap ($c_{i+1} < d_i$), the sum of their lengths telescopes and strictly exceeds $b - a$.
- **Conclusion:** $l(I) < \sum_{i=1}^{k} l(J_i) \leq \sum_{n=1}^{\infty} l(J_n) < m^*(I) + \epsilon$.
- **Other Intervals:** For open or half-open intervals, compress the endpoints by ϵ to create a closed sub-interval, apply the above, and take the limit.

□

Theorem 2.7. *(Countable Subadditivity) Outer measure is countably subadditive, i.e., for any sequence of sets E_n:*

$$m^*\left(\bigcup_{n=1}^{\infty} E_n\right) \leq \sum_{n=1}^{\infty} m^*(E_n)$$

(Note that both sides may be infinite here.)

Crux. The exact same $\epsilon/2^n$ **trick** from Theorem 2.2.

Instead of bounding the cover length by just $\epsilon/2^n$, bound it by the infimum plus the error: for each E_n, choose a cover $\{I_k^n\}_{k\geq 1}$ such that

$$\sum_{k=1}^{\infty} l(I_k^n) < m^*(E_n) + \frac{\epsilon}{2^n}$$

Taking the double sum over all intervals covering $\bigcup E_n$ gives a total length strictly less than $\sum m^*(E_n) + \epsilon$. Since ϵ is arbitrary, the result holds. □

Proposition 2.8 (Translation Invariance) Outer measure is translation-invariant, i.e.,

$$m^*(A) = m^*(A + t)$$

for each $A \subseteq \mathbb{R}$ and $t \in \mathbb{R}$.

Crux. Shifting a cover $\{I_n\}$ of A by t yields a cover $\{I_n + t\}$ of $A + t$. Since interval lengths are unaffected by translation ($l(I_n) = l(I_n + t)$), the sets of all possible covering sums are strictly identical: $Z_A = Z_{A+t}$. Taking the infimum of both sides gives the result at once. □

Definition 2.9 A set $E \subseteq \mathbb{R}$ is (Lebesgue-) measurable if for every set $A \subseteq \mathbb{R}$ we have

$$m^*(A) = m^*(A \cap E) + m^*(A \cap E^c)$$

where $E^c = \mathbb{R} \setminus E$ and we write $E \in \mathcal{M}$.

Theorem 2.10. • *Any null set is measurable.*

- *Any interval is measurable.*

Crux. Recall Carathéodory's condition: E is measurable if $\forall A$, $m^*(A) \geq m^*(A \cap E) + m^*(A \cap E^c)$.

- **Null sets:** Let N be a null set. By monotonicity, $m^*(A \cap N) \leq m^*(N) = 0$, and $m^*(A \cap N^c) \leq m^*(A)$. Adding these gives $0 + m^*(A)$, satisfying the condition.

- **Intervals:** Let I be an interval. Take an ϵ-cover $\{I_n\}$ for A such that $\sum l(I_n) < m^*(A) + \epsilon$. We assume without proof that length is additive for intervals: $l(I_n) = l(I_n \cap I) + l(I_n \cap I^c)$. Using the definition of outer measure for the split covers:

$$m^*(A \cap I) + m^*(A \cap I^c) \leq \sum_{n=1}^{\infty} l(I_n \cap I) + \sum_{n=1}^{\infty} l(I_n \cap I^c) = \sum_{n=1}^{\infty} l(I_n) < m^*(A) + \epsilon$$

  Since this holds $\forall \epsilon > 0$, the result follows.

□

Theorem 2.11.

(i) $\mathbb{R} \in \mathcal{M}$,

(ii) if $E \in \mathcal{M}$ *then* $E^c \in \mathcal{M}$,

(iii) if $E_n \in \mathcal{M}$ *for all* $n = 1, 2, \cdots$ *then* $\cup_{n=1}^{\infty} E_n \in \mathcal{M}$.

Moreover, if $E_n \in \mathcal{M}$, $n = 1, 2, \cdots$ *and* $E_i \cap E_j = \varnothing$ *for* $i \neq j$, *then*

$$m^* \left(\bigcup_{n=1}^{\infty} E_n \right) = \sum_{n=1}^{\infty} m^*(E_n). \tag{2.8}$$

Crux. (i) and (ii) are trivial. The heavy lifting is in (iii) and (2.8).

Phase 1: Disjoint Sets (Recursive Splitting) Assume $\{E_k\}$ are pairwise disjoint. Recursively apply Carathéodory's condition to split a test set A using $E_1, E_2, \ldots, E_n$:

$$\begin{aligned} m^*(A) &= m^*(A \cap E_1) + m^*(A \cap E_1^c \cap E_2) + m^*(A \cap E_1^c \cap E_2^c) \\ &= \sum_{k=1}^{n} m^*(A \cap E_k) + m^* \left(A \cap \left(\bigcup_{k=1}^{n} E_k \right)^c \right) \end{aligned}$$

Phase 2: The Limit Squeeze By monotonicity, $(\bigcup_1^n E_k)^c \supseteq (\bigcup_1^\infty E_k)^c$. Pass the limit $n \to \infty$ on the above equation, then apply countable subadditivity to the sum:

$$\begin{aligned} m^*(A) &\geq \sum_{k=1}^{\infty} m^*(A \cap E_k) + m^*\left(A \cap \left(\bigcup_{k=1}^{\infty} E_k\right)^c\right) \\ &\geq m^*\left(A \cap \left(\bigcup_{k=1}^{\infty} E_k\right)\right) + m^*\left(A \cap \left(\bigcup_{k=1}^{\infty} E_k\right)^c\right) \end{aligned}$$

This single squeeze proves $\bigcup_1^\infty E_k \in \mathcal{M}$. Furthermore, equality holds throughout, so taking $A = \bigcup_1^\infty E_k$ yields countable additivity (2.8).

Phase 3: The Finite Algebra For non-disjoint $E_1, E_2 \in \mathcal{M}$, apply Carathéodory splitting:

$$\begin{aligned} m^*(A) &= m^*(A \cap E_1) + m^*(A \cap E_1^c \cap E_2) + m^*(A \cap E_1^c \cap E_2^c) \\ &\geq m^*(A \cap (E_1 \cup E_2)) + m^*(A \cap (E_1 \cup E_2)^c) \end{aligned}$$

where subadditivity gives $m^*(A \cap E_1) + m^*(A \cap E_1^c \cap E_2) \geq m^*(A \cap (E_1 \cup E_2))$. Thus, $E_1 \cup E_2 \in \mathcal{M}$. By induction, $\mathcal{M}$ is closed under finite unions and complements (and hence set differences).

Phase 4: General Sets (Disjointification) To handle countable non-disjoint $\{E_k\}$, we construct a disjoint sequence: $F_1 = E_1$, $F_2 = E_2 \setminus E_1, \ldots, F_n = E_n \setminus (\cup_{k=1}^{n-1} E_k)$.

Because $\mathcal{M}$ is a finite algebra (Phase 3), each $F_n \in \mathcal{M}$. The F_n are pairwise disjoint, and $\bigcup_1^\infty E_k = \bigcup_1^\infty F_k$. By Phase 2, this countable union is in $\mathcal{M}$. $\square$

Definition 2.12.

- A *σ-field* (or *σ-algebra*) $\mathcal{F}$ on a sample space Ω is a family of subsets containing Ω that is closed under complements and countable unions (properties (i)-(iii) from Theorem 2.11).
- The pair $(\Omega, \mathcal{F})$ is called a *measurable space.*
- A set function $\mu : \mathcal{F} \to [0, \infty]$ is a *measure* if $\mu(\varnothing) = 0$ and it is *countably additive* for pairwise disjoint sets (satisfying (2.8)).
- The triplet $(\Omega, \mathcal{F}, \mu)$ is called a *measure space.*

Note: While often proven specifically for Borel or Lebesgue measures, many of our results naturally extend to any abstract measure space $(\Omega, \mathcal{F}, \mu)$.

Proposition 2.13 If $E_k \in \mathcal{M}$, $k = 1, 2, \cdots$, then

$$E = \bigcap_{k=1}^{\infty} E_k \in \mathcal{M}.$$

Crux. Immediate from De Morgan's laws ($\bigcap_{k=1}^{\infty} E_k = (\bigcup_{k=1}^{\infty} E_k^c)^c$) along with Theorem 2.11 ($\mathcal{M}$ is closed under complements and countable unions). □

Definition 2.14 We shall write $m(E)$ instead of $m^*(E)$ for any $E \in \mathcal{M}$ and call $m(E)$ the Lebesgue measure of the set E.

Proposition 2.15 Suppose $A, B \in \mathcal{M}$.

(i) If $A \subseteq B$, then $m(A) \leq m(B)$.

(ii) If $A \subseteq B$ and $m(A)$ is finite, then $m(B \setminus A) = m(B) - m(A)$.

(iii) m is translation-invariant.

Crux. (i) Immediate from Proposition 2.5, as m is simply m^* restricted to sets in $\mathcal{M}$, and Proposition 2.5 holds for all subsets of $\mathbb{R}$.

(ii) $B \setminus A$ and A are disjoint measurable sets that partition B. By finite additivity (Theorem 2.11), $m(B \setminus A) + m(A) = m(B)$. Subtract the finite value $m(A)$ to yield the result.

(iii) Immediate from Proposition 2.8.

□

Proposition 2.16 If $A \in \mathcal{M}$ and $m(A \Delta B) = 0$, then $B \in \mathcal{M}$ and $m(A) = m(B)$.

Crux. Use the symmetric difference $A \Delta B = (A \setminus B) \cup (B \setminus A)$.

- **Null Subsets:** Since $A \Delta B$ is a null set, its subsets $A \setminus B$ and $B \setminus A$ are also null sets, and therefore measurable.
- **Intersection:** $A \cap B = A \setminus (A \setminus B)$. Since A and $A \setminus B$ are measurable, $A \cap B \in \mathcal{M}$. Since $A \setminus B$ is a null set, additivity gives $m(A) = m(A \setminus B) + m(A \cap B) = 0 + m(A \cap B)$.
- **Reconstruction:** $B = (B \setminus A) \cup (A \cap B)$. Being a union of two measurable sets, $B \in \mathcal{M}$. By additivity again, $m(B) = m(B \setminus A) + m(A \cap B) = 0 + m(A) = m(A)$.

□

Theorem 2.17.

(i) For any $\epsilon > 0$ and $A \subseteq \mathbb{R}$, we can find an open set O such that

$$A \subseteq O, \quad m(O) \leq m^*(A) + \epsilon.$$

Consequently, for any $E \in \mathcal{M}$, we can find an open set $O \supseteq E$ such that $m(O \setminus E) < \epsilon$.

(ii) For any $A \subseteq \mathbb{R}$, we can find a sequence of open sets O_n such that

$$A \subseteq \bigcap_{n=1}^{\infty} O_n, \quad m\left(\bigcap_{n=1}^{\infty} O_n\right) = m^*(A).$$

Crux. (i) **Open Approximation:** Take an $\epsilon/2$-cover $\{I_n\}$ for A. Expand each I_n into an open interval J_n by adding $\epsilon/2^{n+1}$ length. The union $O = \bigcup J_n$ is an open set containing A, and subadditivity guarantees $m(O) \leq \sum l(J_n) < m^*(A) + \epsilon$.

Measurable Case (Finite $m(E)$): Apply Proposition 2.15 directly: $m(O \setminus E) = m(O) - m(E) \leq \epsilon$.

Measurable Case (Infinite $m(E)$): Partition E into finite chunks $E_n = E \cap (-n, n)$. Find open $O_n \supseteq E_n$ with error $< \epsilon/2^n$. Let $O = \bigcup O_n$. Then $O \setminus E \subseteq \bigcup (O_n \setminus E_n)$, and subadditivity bounds the total error strictly by ϵ.

(ii) G_δ **Construction:** For each integer $n \geq 1$, find an open set $O_n \supseteq A$ such that $m(O_n) \leq m^*(A) + 1/n$. Let $G = \bigcap O_n$. Since $A \subseteq G \subseteq O_n$ for all n, monotonicity forces $m^*(A) \leq m(G) \leq m(O_n) \leq m^*(A) + 1/n$. Letting $n \to \infty$ yields the result.

□

Theorem 2.19. *(Continuity of Measure) Suppose that $A_n \in \mathcal{M}$ for all $n \geq 1$. Then we have:*

(i) if $A_n \subseteq A_{n+1}$ for all n, then

$$m\left(\bigcup_n A_n\right) = \lim_{n\to\infty} m(A_n),$$

(ii) if $A_n \supseteq A_{n+1}$ for all n and $m(A_1) < \infty$, then

$$m\left(\bigcap_n A_n\right) = \lim_{n\to\infty} m(A_n).$$

Crux. (i) **Continuity from Below:** Define $B_1 = A_1$ and $B_n = A_n \setminus A_{n-1}$. The $\{B_n\}$ are pairwise disjoint, and $\bigcup A_n = \bigcup B_n$.

$$\begin{aligned} m\left(\bigcup_k A_k\right) &= \sum_{k=1}^{\infty} m(B_k) \\ &= \lim_{n\to\infty} \sum_{k=1}^{n} m(B_k) \\ &= \lim_{n\to\infty} m(A_n) \end{aligned}$$

as the sum precisely telescopes to $m(A_n)$.

(ii) **Continuity from Above:** Define $B_n = A_1 \setminus A_n$. Since A_n is decreasing, B_n is increasing, allowing us to apply the result from (i).

$$\begin{aligned} m\left(\bigcup_n B_n\right) &= m\left(A_1 \setminus \bigcap_n A_n\right) \\ &= m(A_1) - m\left(\bigcap_n A_n\right) \end{aligned}$$

Applying the limit to the increasing sequence yields $\lim_{n\to\infty} m(A_1 \setminus A_n) = m(A_1) - \lim_{n\to\infty} m(A_n)$. Since $m(A_1) < \infty$, we can safely subtract it and rearrange the terms to obtain the desired result.

□

Theorem 2.21. *The set function m satisfies:*

(i) m is finitely additive, i.e., for pairwise disjoint sets $\{A_i\}$, we have

$$m\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n m(A_i)$$

for each n;

(ii) m is continuous at $\varnothing$, i.e., if (B_n) decreases to $\varnothing$, then $m(B_n)$ decreases to 0.

Crux. (i) **Finite Additivity:** Since m is countably additive (Theorem 2.11), set $A_i = \varnothing$ for all $i > n$. The infinite sum trivially truncates to the finite sum because $m(\varnothing) = 0$.

(ii) **Continuity at $\varnothing$ (ERRATA):** The textbook negligently omits the assumption that $m(B_1) < \infty$. (Counterexample: $B_n = [n, \infty)$ decreases to $\varnothing$, but $m(B_n) = \infty \not\to 0$). Assuming $m(B_1) < \infty$, apply Theorem 2.19(ii) (Continuity from Above) directly with $\bigcap_n B_n = \varnothing$ to obtain the limit 0.

□

Theorem 2.22. *The intersection of any family of σ-fields is a σ-field.*

Crux. Let $\mathcal{F} = \bigcap_{\alpha\in\Lambda} \mathcal{F}_\alpha$, where each $\mathcal{F}_\alpha$ is a σ-field. We verify the three axioms for $\mathcal{F}$:

- **Whole Space:** Since $\mathbb{R} \in \mathcal{F}_\alpha$ for every $\alpha \in \Lambda$, it follows that $\mathbb{R} \in \mathcal{F}$.
- **Complements:** If $E \in \mathcal{F}$, then $E \in \mathcal{F}_\alpha$ for all α. Since each $\mathcal{F}_\alpha$ is a σ-field, $E^c \in \mathcal{F}_\alpha$ for all α, which implies $E^c \in \mathcal{F}$.

- **Countable Unions:** If $E_k \in \mathcal{F}$ for $k \geq 1$, then $E_k \in \mathcal{F}_\alpha$ for all $k \geq 1$ and for every α. Since each $\mathcal{F}_\alpha$ is closed under countable unions, $\bigcup_{k=1}^\infty E_k \in \mathcal{F}_\alpha$ for every α. Therefore, $\bigcup_{k=1}^\infty E_k \in \mathcal{F}$.

$\square$

Definition 2.23 Put

$$\mathcal{B} = \bigcap \{\mathcal{F} : \mathcal{F} \text{ is a } \sigma\text{−field containing all intervals}\}.$$

We say that $\mathcal{B}$ is the σ-field generated by all intervals and we call the elements of $\mathcal{B}$ *Borel* sets.

Theorem 2.25. *If instead of the family of all intervals we take all open intervals, all closed intervals, all intervals of the form (a, ∞) (or of the form $[a, \infty)$, $(-\infty, b)$, or $(-\infty, b]$), all open sets, or all closed sets, then the σ-field generated by them is the same as $\mathcal{B}$.*

Crux. We have the Borel σ-field defined as

$$\mathcal{B} = \bigcap \{\mathcal{F} : \mathcal{F} \text{ is a } \sigma\text{-field containing all intervals } \mathcal{I}\}.$$

Let $\mathcal{C}$ be the σ-field generated by open intervals:

$$\mathcal{C} = \bigcap \{\mathcal{F} : \mathcal{F} \text{ is a } \sigma\text{-field containing all open intervals } \mathcal{J}\}.$$

- **Showing $\mathcal{C} \subseteq \mathcal{B}$:** If a σ-field contains all intervals $\mathcal{I}$, it trivially contains all open intervals $\mathcal{J}$. Therefore, the collection of σ-fields $\{\mathcal{F} : \mathcal{F} \supseteq \mathcal{I}\} \subseteq \{\mathcal{F} : \mathcal{F} \supseteq \mathcal{J}\}$. Taking the intersection of these collections reverses the inclusion, yielding $\bigcap\{\mathcal{F} : \mathcal{F} \supseteq \mathcal{J}\} \subseteq \bigcap\{\mathcal{F} : \mathcal{F} \supseteq \mathcal{I}\}$, which means $\mathcal{C} \subseteq \mathcal{B}$.

- **Showing $\mathcal{B} \subseteq \mathcal{C}$:** Any interval can be expressed as a countable intersection of open intervals. For example:

$$[a, b) = \bigcap_{n=1}^{\infty} \left(a - \frac{1}{n}, b\right)$$

  Since $\mathcal{C}$ is a σ-field containing all open intervals, it must also contain all countable intersections of them. Thus, $\mathcal{C} \supseteq \mathcal{I}$. Because $\mathcal{B}$ is by definition the *smallest* σ-field containing $\mathcal{I}$, we must have $\mathcal{B} \subseteq \mathcal{C}$.

Hence, $\mathcal{B} = \mathcal{C}$. $\square$

Definition: Complete Measure Space – A measure space $(X, \mathcal{F}, \mu)$ is *complete* if for all $F \in \mathcal{F}$ with $\mu(F) = 0$, for all $N \subset F$ we have $N \in \mathcal{F}$ (and so $\mu(N) = 0$).

Definition: Completion of a σ-field – The *completion* of a σ-field $\mathcal{G}$, relative to a given measure μ, is defined as the smallest σ-field $\mathcal{F}$ containing $\mathcal{G}$ such that, if $N \subset G \in \mathcal{G}$ and $\mu(G) = 0$, then $N \in \mathcal{F}$.

Proposition 2.27

ERRATA: Incorrect Textbook Statement – The completion of $\mathcal{G}$ is of the form $\{G \cup N : G \in \mathcal{F}, N \subset F \in \mathcal{F} \text{ with } \mu(F) = 0\}$.

Correct Statement – The completion $\mathcal{F}$ of a σ-field $\mathcal{G}$ is of the form $\{G \cup N : G \in \mathcal{G}, N \subset F \in \mathcal{G} \text{ with } \mu(F) = 0\}$.

Crux. Let $\mathcal{K} \triangleq \{G \cup N : G \in \mathcal{G}, N \subset F \in \mathcal{G} \text{ with } \mu(F) = 0\}$. We need to show that $\mathcal{K} = \mathcal{F}$.

Phase 1: $\mathcal{K}$ is a σ-field

- **Whole Space:** $\mathbb{R} = \mathbb{R} \cup \varnothing$. Since $\mathbb{R} \in \mathcal{G}$ and $\varnothing \subset \varnothing \in \mathcal{G}$ with $\mu(\varnothing) = 0$, we have $\mathbb{R} \in \mathcal{K}$.
- **Countable Unions:** Let $G_k \cup N_k \in \mathcal{K}$, with $N_k \subset F_k \in \mathcal{G}$ where $\mu(F_k) = 0$. Since $\mathcal{G}$ is a σ-field, $\bigcup_k G_k \in \mathcal{G}$ and $\bigcup_k F_k \in \mathcal{G}$. By subadditivity, $\mu(\bigcup_k F_k) = 0$. Since $\bigcup_k N_k \subset \bigcup_k F_k$, the union $\bigcup_k (G_k \cup N_k) \in \mathcal{K}$.
- **Complements (The Partition Trick):** Let $G \cup N \in \mathcal{K}$. Express the null bound F as the partition $F = N \cup M$, where $M = F \setminus N$.

  $$(G \cup N)^c = G^c \cap N^c = G^c \cap (F \setminus M)^c = G^c \cap (F^c \cup M) = (G \cup F)^c \cup (G^c \cap M)$$

  Since $G, F \in \mathcal{G}$, the base $(G \cup F)^c \in \mathcal{G}$. The remainder $(G^c \cap M) \subset F \in \mathcal{G}$ where $\mu(F) = 0$. Thus, $(G \cup N)^c \in \mathcal{K}$.

Phase 2: Minimality ($\mathcal{K} = \mathcal{F}$) $\mathcal{K}$ is a σ-field containing $\mathcal{G}$ (let $N = \varnothing$) and all subsets of null sets of $\mathcal{G}$ (let $G = \varnothing$). Since $\mathcal{F}$ is defined as the *smallest* such σ-field, $\mathcal{F} \subseteq \mathcal{K}$. Conversely, any σ-field containing $\mathcal{G}$ and its null subsets must, by closure under finite unions, contain all sets of the form $G \cup N$. Therefore $\mathcal{K} \subseteq \mathcal{F}$, proving $\mathcal{F} = \mathcal{K}$. □

Theorem 2.28. *$\mathcal{M}$ is the completion of $\mathcal{B}$.*

Crux. Let $\mathcal{K} \triangleq \{B \cup N : B \in \mathcal{B}, N \subseteq F \in \mathcal{B} \text{ with } m(F) = 0\}$ be the completion of $\mathcal{B}$. We need to show $\mathcal{K} = \mathcal{M}$.

Phase 1: $\mathcal{K} \subseteq \mathcal{M}$ We know $\mathcal{M}$ is a σ-field and contains $\mathcal{B}$. Let $N \subseteq F \in \mathcal{B}$ with $m(F) = 0$. By monotonicity (Proposition 2.5), $m^*(N) \leq m^*(F) = m(F) = 0$, so $m^*(N) = 0$. By Theorem 2.10, all null sets are Lebesgue measurable, so $N \in \mathcal{M}$. Since $\mathcal{M}$ is closed under unions, $B \cup N \in \mathcal{M}$. Because $\mathcal{K}$ is defined exactly as the collection of all such unions, $\mathcal{K} \subseteq \mathcal{M}$.

Phase 2: $\mathcal{M} \subseteq \mathcal{K}$ Let $E \in \mathcal{M}$.

- **Finite Measure Case:** Assume $m^*(E) < \infty$. By outer regularity (Theorem 2.17), we can find a G_δ set $B = \bigcap_n O_n \supseteq E$ (where O_n are open, hence $B \in \mathcal{B}$) such that $m(B) = m^*(E)$. Let $N \triangleq B \setminus E$. Since $m^*(E)$ is finite, $m^*(N) = m(B) - m^*(E) = 0$, meaning $N \in \mathcal{M}$. Applying Theorem 2.17 again to N, we find a Borel set $L \supseteq N$ such that $m(L) = m^*(N) = 0$. Thus, N is a subset of a Borel null set L, meaning $N \in \mathcal{K}$ (specifically, $N = \varnothing \cup N$). Since $B \in \mathcal{B} \subseteq \mathcal{K}$ and $N \in \mathcal{K}$, and $\mathcal{K}$ is a σ-field (Proposition 2.27), it follows that $E = B \setminus N \in \mathcal{K}$.
- **Infinite Measure Case:** Assume $m^*(E) = \infty$. Truncate the set by defining $E_n = [-n, n] \cap E$. By the finite case above, each $E_n \in \mathcal{K}$. Since $\mathcal{K}$ is a σ-field, the countable union $E = \bigcup_n E_n \in \mathcal{K}$.

Therefore, $\mathcal{M} \subseteq \mathcal{K}$, concluding that $\mathcal{M} = \mathcal{K}$. □

Theorem 2.29. *If $E \in \mathcal{M}$, then for any given $\epsilon > 0$ there exists a closed set $F \subseteq E$ such that $m(E \setminus F) < \epsilon$. Hence, there exists a set $B \subseteq E$ of the form $B = \bigcup_n F_n$ (where all F_n are closed sets) such that $m(E \setminus B) = 0$.*

***Crux*.** By duality, we apply outer regularity (Theorem 2.17) to the complement $E^c \in \mathcal{M}$.

- **Closed Approximation:** There exists an open set $O \supseteq E^c$ such that $m(O \setminus E^c) < \epsilon$. Observe that $O \setminus E^c = O \cap (E^c)^c = O \cap E = E \setminus O^c$. Let $F = O^c$. Since O is open, F is closed. Since $O \supseteq E^c$, taking complements gives $F = O^c \subseteq E$. Thus, $m(E \setminus F) < \epsilon$.
- **F_σ Construction:** By Theorem 2.17(ii), we can find a G_δ set $G = \bigcap_n O_n \supseteq E^c$ (where O_n are open sets) such that $m(G \setminus E^c) = 0$. Taking complements, let $B = G^c = \bigcup_n O_n^c$. Let $F_n = O_n^c$, which are closed, making B an F_σ set. Since $G \supseteq E^c$, we have $B \subseteq E$. Finally, $m(E \setminus B) = m(E \cap G) = m(G \setminus E^c) = 0$.

□

Exercise 2.8 Show that each of the following two statements is equivalent to saying that $E \in \mathcal{M}$:

(i) given $\epsilon > 0$ there is an open set $O \supset E$ with $m^*(O \setminus E) < \epsilon$,

(ii) given $\epsilon > 0$ there is a closed set $F \subset E$ with $m^*(E \setminus F) < \epsilon$.

***Crux*.** (i) **Open Approximation:** Assume for a set E and any $\epsilon > 0$, we can find an open set $O \supseteq E$ such that $m^*(O \setminus E) < \epsilon$. Let $N \triangleq O \setminus E$, so that $E^c = O^c \cup N$, with $m^*(N) < \epsilon$. Since open sets are measurable, $O \in \mathcal{M}$. Testing E via Carathéodory's criterion and using subadditivity:

$$\begin{aligned} m^*(A \cap E) + m^*(A \cap E^c) &= m^*(A \cap E) + m^*((A \cap O^c) \cup (A \cap N)) \\ &\leq m^*(A \cap O) + m^*(A \cap O^c) + m^*(N) \\ &= m^*(A) + m^*(N) \\ &< m^*(A) + \epsilon \end{aligned}$$

Since this holds for any $\epsilon > 0$, we have $m^*(A \cap E) + m^*(A \cap E^c) \leq m^*(A)$, proving $E \in \mathcal{M}$.

(ii) **Closed Approximation (Duality):** Apply (i) to E^c. The closed set $F \subseteq E$ yields an open set $O = F^c \supseteq E^c$. Since $O \setminus E^c = E \setminus F$, the condition $m^*(E \setminus F) < \epsilon$ translates to $m^*(O \setminus E^c) < \epsilon$. By (i), $E^c \in \mathcal{M}$, which implies $E \in \mathcal{M}$.

□

Proposition 2.31 Let $B \in \mathcal{M}$. Given Lebesgue measure m on the Lebesgue σ-field $\mathcal{M}$, let $\mathcal{M}_B = \{A \cap B : A \in \mathcal{M}\}$ and for $A \in \mathcal{M}_B$, write $m_B(A) = m(A)$. Then $(B, \mathcal{M}_B, m_B)$ is a complete measure space.

Crux.

- $\mathcal{M}_B$ **is a σ-field on B:** Whole space: $B = B \cap B \in \mathcal{M}_B$. Complements: $B \setminus (A \cap B) = A^c \cap B \in \mathcal{M}_B$. Countable Unions: $\bigcup_n (A_n \cap B) = (\bigcup_n A_n) \cap B \in \mathcal{M}_B$.

- m_B **is a measure:** Inherits countable additivity trivially from m, since disjoint sets in $\mathcal{M}_B$ are disjoint sets in $\mathcal{M}$.

- **Completeness:** Let $N \subseteq F \in \mathcal{M}_B$ with $m_B(F) = 0$. Since $F = A \cap B$ for some $A \in \mathcal{M}$, we have $F \in \mathcal{M}$ and $m(F) = 0$. The completeness of $\mathcal{M}$ guarantees $N \in \mathcal{M}$. Because $N \subseteq F \subseteq B$, we can write $N = N \cap B$, proving $N \in \mathcal{M}_B$.

□

Definition 2.32 A *probability space* is a triple $(\Omega, \mathcal{F}, P)$ where Ω is an arbitrary set, $\mathcal{F}$ is a σ-field of subsets of Ω, and P is a measure on $\mathcal{F}$ such that

$$P(\Omega) = 1,$$

called *probability measure* or, briefly, *probability*.

Definition 2.35 Given a probability space $(\Omega, \mathcal{F}, P)$ we say that the elements of $\mathcal{F}$ are *events*.

Definition 2.36 Suppose that $P(B) > 0$. Then the number

$$P(A|B) \triangleq \frac{P(A \cap B)}{P(B)}$$

is called the *conditional probability* of A *given* B.

Proposition 2.37 The mapping $A \mapsto P(A|B)$ is countably additive on the σ-field $\mathcal{F}$.

Crux. Let $\{A_n\}$ be a sequence of pairwise disjoint sets in $\mathcal{F}$. Consequently, the sets $\{A_n \cap B\}$ are also pairwise disjoint in $\mathcal{F}$. Distributing the intersection and applying the countable additivity of P:

$$\begin{aligned} P\left(\bigcup_n A_n \middle| B\right) &= \frac{P\left(\left(\bigcup_n A_n\right) \cap B\right)}{P(B)} \\ &= \frac{P\left(\bigcup_n (A_n \cap B)\right)}{P(B)} \\ &= \sum_n \frac{P(A_n \cap B)}{P(B)} \\ &= \sum_n P(A_n|B) \end{aligned}$$

$\square$

Exercise 2.9 Prove that if H_i are pairwise disjoint events such that $\cup_{i=1}^{\infty} H_i = \Omega$, $P(H_i) \neq 0$, then

$$P(A) = \sum_{i=1}^{\infty} P(A|H_i)P(H_i).$$

Crux. The sets $\{A \cap H_i\}$ are pairwise disjoint and their union is A. Hence $P(A) = \sum_{i=1}^{\infty} P(A \cap H_i) = \sum_{i=1}^{\infty} P(A|H_i)P(H_i)$ using Definition 2.36. $\square$

Definition 2.38 The events A, B are *independent* if

$$P(A \cap B) = P(A)P(B)$$

.

Definition 2.40 The events $A_1, \cdots, A_n$ are independent if for all $k \leq n$ for each choice of k events, the probability of their intersection is the product of the probabilities.

Definition 2.41 The σ-fields $\mathcal{F}_1, \mathcal{F}_2, \cdots, \mathcal{F}_n$ defined on a probability space $(\Omega, \mathcal{F}, P)$ are *independent* if, for all choices of distinct indices $i_1, i_2, \cdots, i_k$ from $\{1, 2, 3, \cdots, n\}$ and all choices of sets $F_{i_l} \in \mathcal{F}_{i_l}$ with l from $\{1, 2, 3, \cdots, k\}$, we have

$$P(F_{i_1} \cap F_{i_2} \cap \cdots F_{i_k}) = P(F_{i_1})P(F_{i_2}) \cdots P(F_{i_k})$$

.